Paper id: 252836

Roll No: | | | | | | | | | | | | | |

BPHARM

(SEM I) THEORY EXAMINATION 2024-25

PHARMACEUTICAL INORGANIC CHEMISTRY– THEORY

TIME: 3 HRS **M.MARKS: 75**

Note: 1. Attempt all Sections. If require any missing data; then choose suitably.

SECTION A

1. Attempt *all* questions in brief. 10 x 2 = 20

a.	What is radioactivity? Give the unit of radioactivity.
b.	Define impurities in pharmaceutical substances.
c.	Write the Henderson-Hasselbalch equation.
d.	Name two ideal properties of antacids.
e.	Mention the role of potassium iodide as an expectorant.
f.	State the antidote action of sodium thiosulphate.
g.	What are isotonic solutions? Give one example.
h.	Write two medicinal uses of calcium gluconate.
i.	State the use of boric acid as an antimicrobial.
j.	Define half-life. Mention its importance in radiopharmaceuticals.

SECTION B

2. Attempt any *two* parts of the following: 2 x 10 = 20

a.	Explain in detail the sources and types of impurities in pharmaceutical substances.
b.	Define antacids. Explain the ideal properties of antacids, and write a note on aluminum hydroxide gel and magnesium hydroxide mixture.
c.	Write a detailed note on radioactivity: definition, types of radiation, and measurement techniques.

SECTION C

3. Attempt any *five* parts of the following: 7 x 5 = 35

a.	What is a buffer? Explain buffer equations, buffer capacity, and their significance in pharmacy.
b.	Discuss the hematinic used in pharmaceutical preparations with emphasis on ferrous sulphate and ferrous gluconate.
c.	Write a short note on expectorants and emetics, with examples like ammonium chloride and copper sulphate.
d.	Write a note on acidifiers with emphasis on ammonium chloride and dilute HCl.
e.	Describe the composition, use, and mechanism of action of Oral Rehydration Salt (ORS).
f.	Write short notes on: a)Magnesium hydroxide mixture b) Sodium bicarbonate c) Kaolin and Bentonite
g.	Write a short note on the limit test for iron and arsenic with chemical reactions

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